

Maths Unit 6, Part 2: Radian, Arc aur Sector

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Pichhle safhay par hum ne **degree, minute aur second** seekhe thay. Aaj ek bilkul naya paimana hai: **radian**. Pehle ek sawal: zaviya naapne ka ek tareeqa to pehle se mojud hai, phir doosra kyun banaya gaya?

1 Radian kya hai

Ek circle lein jis ka **nasf qutar (radius) r** hai. Us ke kinare par ek aisa **arc** kaatein jis ki **lambai bilkul r ke barabar** ho.

Wo arc markaz par jo zaviya banata hai, **usi ka naam 1 radian hai**.

Yani radian ko kisi bahar ki cheez se nahin, **circle ki apni radius** se naapa jata hai. Yehi is ki khoobi hai.

Degree aur radian ka pul

π radians = 180° (yani 2π radians = poora 360°)

Is ek line se dono taraf ka kaam nikal aata hai:

Degree se radian: $\times \pi/180$ **Radian se degree:** $\times 180/\pi$

Yaad rakhne ka tareeqa: jahan jana hai wo **ooper** likhein. Radian chahiye to π ooper, degree chahiye to 180 ooper.

Dono taraf ki misalein

45° ko radian mein: $45 \times \pi/180 = \pi/4$

150° ko radian mein: $150 \times \pi/180 = 5\pi/6$

$7\pi/40$ ko degree mein: $(7/40) \times 180 = 1260/40 = 31.5^\circ = 31^\circ 30' 0''$

Aur 1 radian taqreeban **$57^\circ 17' 45''$** hota hai. Ye number yaad rakhein.

6.2 Sector of Circle: do formulay

Kya nikalna hai	Formula	Yaad rakhne ki baat
Arc ki lambai	$l = r\theta$	Kitab: "length of a circular arc is product of radius and radian measure of central angle"
Sector ka raqba	$A = \frac{1}{2} r^2 \theta$	Yani $A = r^2 \theta / 2$. Ye circle ke raqbe πr^2 ka wo hissa hai jo θ ghairta hai

⚠ Sab se bari ghalti, aur wo yehi hai

In dono formulon mein **θ hamesha RADIAN mein** hona chahiye. **Degree bilkul nahin.**

Agar sawal mein zaviya degree mein diya ho to **pehle radian mein badlein**, phir formula lagayein. Jo log seedha degree daal dete hain un ka jawab hamesha ghalat aata hai.

✏ Ek mukammal misal

$r = 10$ cm, $\theta = 2\pi/5$ radians.

Arc: $l = r\theta = 10 \times 2\pi/5 = 4\pi \approx 12.57$ cm

Raqba: $A = \frac{1}{2} \times 10^2 \times 2\pi/5 = \frac{1}{2} \times 100 \times 2\pi/5 = 20\pi \approx 62.83$ cm²

Unit ka dhyan: lambai **cm** mein, raqba **cm²** mein. Unit na likhne par aadha mark jata hai.

✏ Quiz

- Define **one radian** in the words of your book, write the relation between radians and degrees, and convert **150°** into radians in terms of π .
- A circle has radius **14 cm** and a sector whose arc length is **22 cm**. Find the central angle in radians, then the area of that sector. Give the unit with each answer.